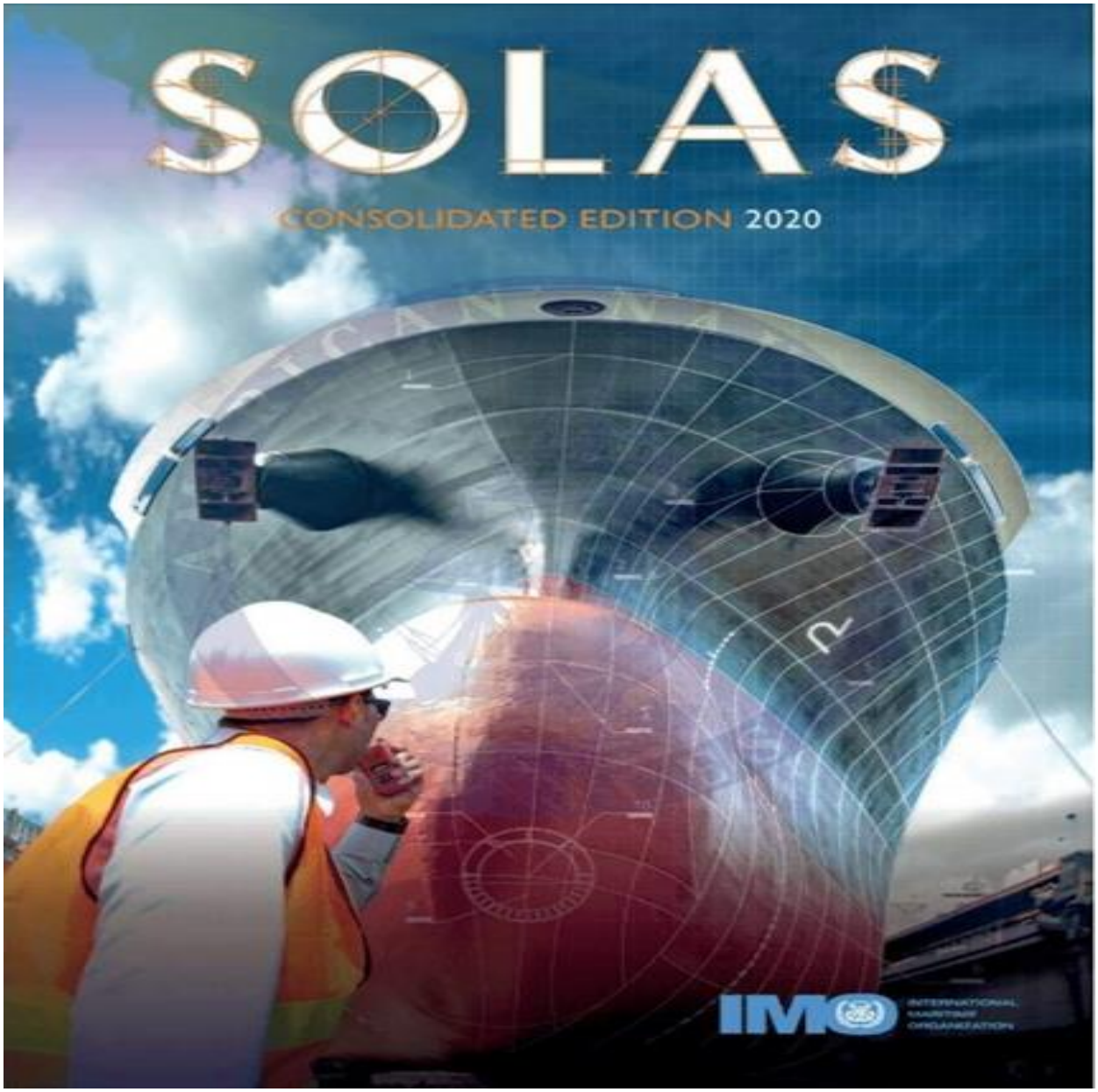
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DEPARTMENT OF MARINE ENGINEERING

REPORT OF VALUE ADDED COURSE



Value Added Course

VAC23003-Safety of Life at Sea (SOLAS)

PSN College of Engineering and Technology
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DEPARTMENT OF MARINE ENGINEERING

22.11.2023

Submitted to Principal for kind approval

Respected sir,

The Department of Marine Engineering is organizing the Value-Added Course during the academic year 2023-24 (ODD).

We submit this for your kind approval

S.No	Name of VAC Programme	Course Coordinator	Resource person	Duration (5 Days) (27/11/2023-01/12/2023)	Total hours
1	**Safety of Life at Sea (SOLAS)** (VAC23003)	Mr.S.Ramkumar, Asst. Prof/Marine	Er.S.Manivannan, Marine Engineer Officer	10.00 am – 1.00 pm 2.00 pm – 4.00 pm	30

22/11/23

HOD/MARINE

Principal

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DEPARTMENT OF MARINE ENGINEERING

PSNCET/MARINE/2023-24/VAC/II Yr. **Date: 22.11.2023**

CIRCULAR

The Department of MARINE is organizing the following Value-Added Course from 27.11.2023 to 01.12.2023. All the MARINE department students and other willing students are informed to register their names to respective Course Coordinator on or before 24.11.2023. The two credits will be awarded to the participants who have successfully completed the Value-Added Course.

S.No	Name of VAC Programme	Course Coordinator	Resource person	Duration (27/11/2023-01/12/2023)	Total hours
1	**Safety of Life at Sea (SOLAS)** (VACZ3003)	Mr.S.Ramkumar, Asst. Prof/Marine	Er.S.Manivannan, Marine Engineer Officer	10.00 am – 1.00 pm 2.00 pm – 4.00 pm	30

22/11/23

HOD/MARINE

Copy to:

1. The Principal, PSNCET
2. IQAC, PSNCET
3. Dept. Notice Board

FIVE DAYS VALUE ADDED COURSE

ON

"Safety of Life at Sea (SOLAS)"

(27.11.2023 to 01.12.2023)

Organized by Department of Marine Engineering

PSN College of Engineering and Technology

(An Autonomous Institution)

(Affiliated to Anna University, Chennai, Approved by AICTE, Recognized by UGC under section 2(f), An ISO 9001:2015 Certified Institution, Accredited by NAAC.

Melathediyoor, Tirunelveli - 627 152

www.psncet.ac.in

ABOUT THE INSTITUTE

College was established by the PSN Educational and Charitable Trust in the year 2001 under the leadership of our illustrious Chairman Dr.P.Suyambu, Ph.D, a multi-tasked personality and an admirable academic brilliance. The college is located at Melathediyoor, 12 km away from Tirunelveli unfolding its grandeur over 250 acres of land.

The college offers 9 UG, 7 PG and 5 Ph.D. programmes. On campus academic and Research activities drive both faculty and students towards their goals with the end results of research publications, funded projects, academic achievements and placement offers.

Vision

Provide an academic environment to learn, work and do research and enabling the students to face challenges in life with strong ethical values.

Mission

- To achieve greater heights of excellence in technical knowledge and skill development through innovative teaching and learning practices.
- To develop the infrastructure to meet the demands of technological revolution.
- To improve and foster research in all dimensions for betterment of society.
- To develop individual competencies to enhance employability and entrepreneurship of students.
- To instil higher standards of discipline among students, including ethical and moral values for societal harmony and peace.

ABOUT EEE TECHNICAL FORUM

Department of Marine has started technical forum in the name **STREAMARO** and effectively organizing technical events like guest lecturers, workshops and faculty development programs to the faculty members. Since the academic year 2002 -2003, it has joined hands to increase the number of events and the quality in the events. The forum executive body actively plans and executes the programs. This time it brings up a value added course for the benefit of the students and staff.

Department Vision

To Produce competent, disciplined quality Engineers and administrators in accordance with global standards in Marine Engineering with capability of accepting new challenges.

Department Mission

To develop and train the students in order to meet dynamic needs and demands of the society.

To promote research and continuing education.

To enhance professional and entrepreneurial skills with ethical values through industry - institute interaction.

OBJECTIVES OF THE VALUE ADDED COURSE

This five days value added course aims how to ensure awareness regarding Environmental Protection at Sea and to impact aspect of commitment.

RESOURCE PERSONS:

Er.S.Manivannan, Marine Engineer Officer.

PARTICIPANTS

Students of Marine Departments.

REGISTRATION

The participants are requested to confirm their registration to the coordinator. Spot registration facility is also available. Registration is on first–come–first–serve basis and registrations are limited.

REGISTRATION FEE

No Registration fee

IMPORTANT DATES

Last date for receipt of registration form – **24.11.2023**

CERTIFICATE

A **certificate** of participation will be issued.

Chief patron:

Dr. P. Suyambu, Chairman.

Patron:

Er. P.S. Jayaram, Vice Chairman.

Co-Patron(s)

Dr.P.Selvakumar, Executive Director

Dr. V.Manikandan, Principal.

Convener

Dr. R.Satheesh Raja, HoD/Marine.

Coordinator

Mr.S.Ramkumar, Asst. Prof/Marine

Contact Details

Feel free to ask your queries either by Email: hodmarine@psncet.ac.in

Contact: - 9487303805

REGISTRATION FORM

Reg. ID (for office use only):			

Personal Details:

Name :

Designation :

(Student must append your roll number)

College Name :

College Address :

Branch :

(Student must append their year of study)

Communication Details:

Mobile :

E-Mail :

Signature of HOD / PRINCIPAL

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DEPARTMENT OF MARINE ENGINEERING

Academic Year 2023-2024

Value Added course on "VAC23003-Safety of Life at Sea (SOLAS)"

Course Content

Total: 30 Hours

Module 1: International Treaties, Conventions, Protocols, Rules, Legislations and Regulations and Law of the Sea 1982 (UNCLOS)
(i) Conventions
(ii) Amendments

Module 2: International Maritime Organisation (IMO) 1948
(i) Organs of the IMO
(ii) Main committees

Module 3: SOLAS Convention 1974
(i) Contents of SOLAS
(ii) SOLAS- Chapter

Module 4: Ship Certificates / Documents under SOLAS
(i) Documents to be carried on board
(ii) Certificates and Documents required on board as per ILO Convention

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Module 5: Regulatory Surveys and Penalties
(i) Validity of certificates
(ii) Regulatory certificates and penalties

Head of the Department / MARINE

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Melathediyoor, Palayamkottai, Tirunelveli – 627 152
DEPARTMENT OF MARINE ENGINEERING

Academic Year 2023-2024

Value Added course on "VAC23003-Safety of Life at Sea (SOLAS)"

Course Plan

Course	Value Added Course
Course Code	VAC23003
Title of the Course	Safety of Life at Sea (SOLAS)
Dates & Duration	27.11.2023 to 01.12.2023 & 30 Hours

S.No.	Topic to be covered	Hours Needed	Mode of Teaching (PPT/Others)
International Treaties, Conventions, Protocols, Rules, Legislations and Regulations and Law of the Sea 1982 (UNCLOS)			
1	Conventions	3	PPT
2	Amendments	3	PPT
International Maritime Organization (IMO) 1948			
3	Organs of the IMO	3	PPT
4	Main committees	3	PPT
SOLAS Convention 1974			
5	Contents of SOLAS	3	PPT
6	SOLAS- Chapter	3	PPT
Ship Certificates / Documents under SOLAS			
7	Documents to be carried on board	3	PPT
8	Certificates and Documents required on board as per ILO Convention	3	PPT
Regulatory Surveys and Penalties			
9	Validity of certificates	3	PPT
10	Regulatory certificates and penalties	2	PPT
11	Test and Feedback	1	

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Head of the Department / MARINE

PSN COLLEGE OF ENGINEERING AND TECHNOLOGY
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Accredited By NAAC with A+
Melathediyoor, Palayamkottai, Tirunelveli – 627 152

DEPARTMENT OF MARINE ENGINEERING

25.11.2023

List of students registered for Value Added Course on "Safety of Life at Sea (SOLAS)"

S.NO	ROLL NO.	NAME OF THE STUDENT	Year of study
1	95222208001	ANANDHA NARAYANAN M	II
2	95222208002	ANUSIYA B	II
3	95222208003	ARAVIND V	II
4	95222208004	ASHOKKUMAR M	II
5	95222208005	BASKAR A	II
6	95222208006	ESWARAN T	II
7	95222208007	SAJIL KUMAR P	II
8	95222208008	SANJAI S	II
9	95222208009	SATHEESH KUMAR C	II
10	95222208010	SOMA SUNDARAM P	II
11	95222208011	SUNDARA PANDI S	II
12	95222208012	TAMIL SELVAN A	II
13	95222208013	THRANESHWARAN A	II
14	95222208014	VAITHEESWARAN K	II
15	95222208015	VIJAYARAJAPANDI K	II
16	95222208016	VISHWANATH N	II
17	95222208301	JOHN AVINASH M	II
18	95222208302	MUTHU SAMY K	II

Head of the Department / MARINE

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DEPARTMENT OF MARINE ENGINEERING

Value Added Course on "VAC23003-Safety of Life at Sea (SOLAS)"

Attendance Sheet

DATE: 27.11.2023 to 01.12.2023

S.NO	ROLL NO.	NAME OF THE STUDENT	27.11.2023	28.11.2023	29.11.2023	30.11.2023	01.12.2023
1	95222208001	ANANDHA NARAYANAN M					
2	95222208002	ANUSIYA B					
3	95222208003	ARAVIND V					
4	95222208004	ASHOKKUMAR M					
5	95222208005	BASKAR A					
6	95222208006	ESWARAN T					
7	95222208007	SAJIL KUMAR P					
8	95222208008	SANJAI S					
9	95222208009	SATHEESH KUMAR C					
10	95222208010	SOMA SUNDARAM P					
11	95222208011	SUNDARA PANDI S					
12	95222208012	TAMIL SELVAN A					
13	95222208013	THRANESHWARAN A					
14	95222208014	VAITHEESWARAN K					
15	95222208015	VIJAYARAJAPANDI K	AB	AB	AB	AB	AB
16	95222208016	VISHWANATH N					
17	95222208301	JOHN AVINASH M					
18	95222208302	MUTHU SAMY K					

VAC Coordinator

Head of the Department / MARINE

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DEPARTMENT OF MARINE ENGINEERING

Value Added Course on "VAC23003-Safety of Life at Sea (SOLAS)" (27.11.2023 to 01.12.2023)

Mark Statement

S.No	Register Number	Name Of The Student	Mark
1	95222208001	ANANDHA NARAYANAN M	93
2	95222208002	ANUSIYA B	80
3	95222208003	ARAVIND V	80
4	95222208004	ASHOKKUMAR M	87
5	95222208005	BASKAR A	93
6	95222208006	ESWARAN T	93
7	95222208007	SAJIL KUMAR P	87
8	95222208008	SANJAI S	87
9	95222208009	SATHEESH KUMAR C	93
10	95222208010	SOMA SUNDARAM P	93
11	95222208011	SUNDARA PANDI S	87
12	95222208012	TAMIL SELVAN A	93
13	95222208013	THRANESHWARAN A	93
14	95222208014	VAITHEESWARAN K	87
15	95222208015	VIJAYARAJAPANDI K	AB
16	95222208016	VISHWANATH N	93
17	95222208301	JOHN AVINASH M	87
18	95222208302	MUTHU SAMY K	93

VAC Coordinator

Head of the Department / MARINE

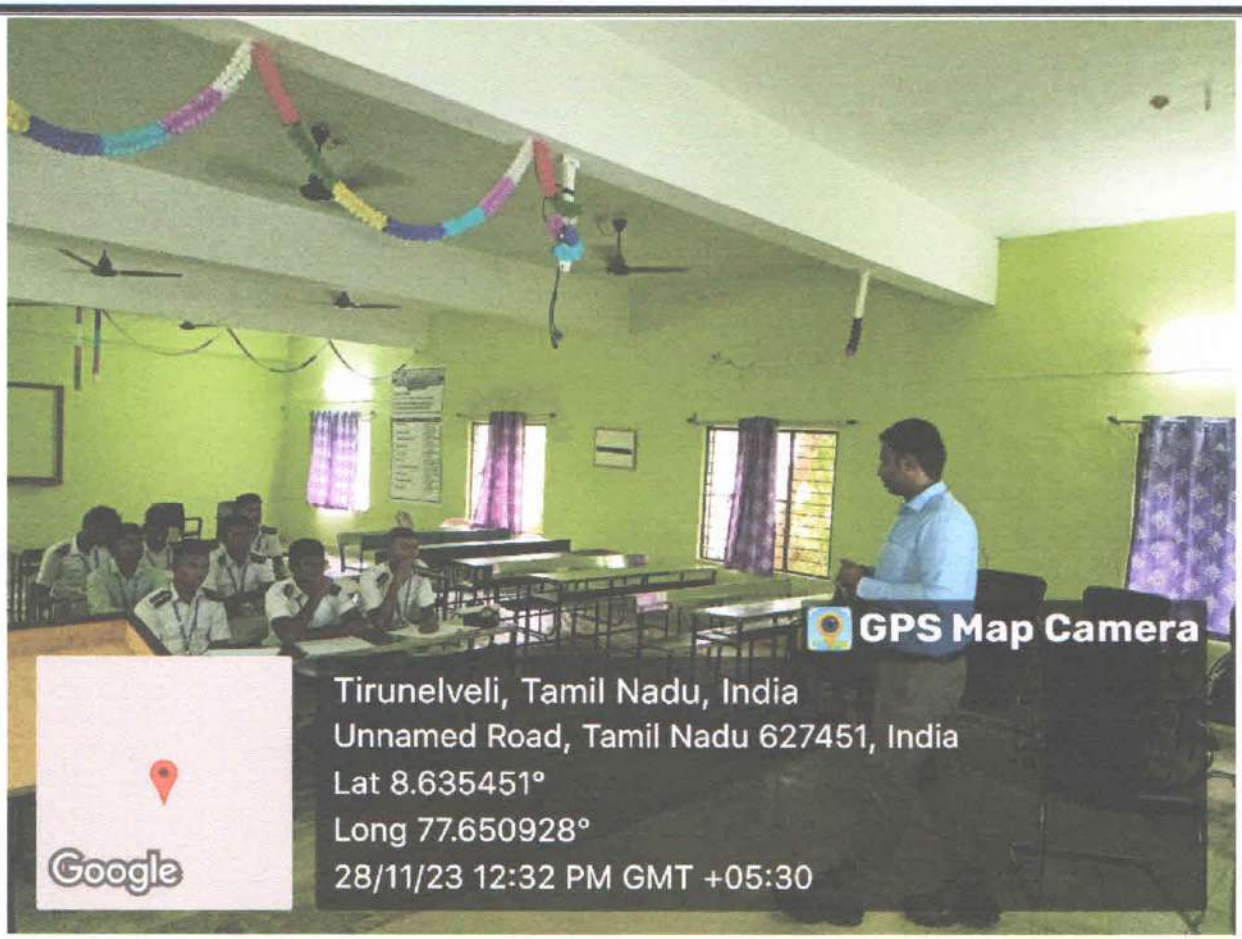
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DEPARTMENT OF MARINE ENGINEERING



Value Added Course on "VAC23003-Safety of Life at Sea (SOLAS)"
(27.11.2023 to 01.12.2023)

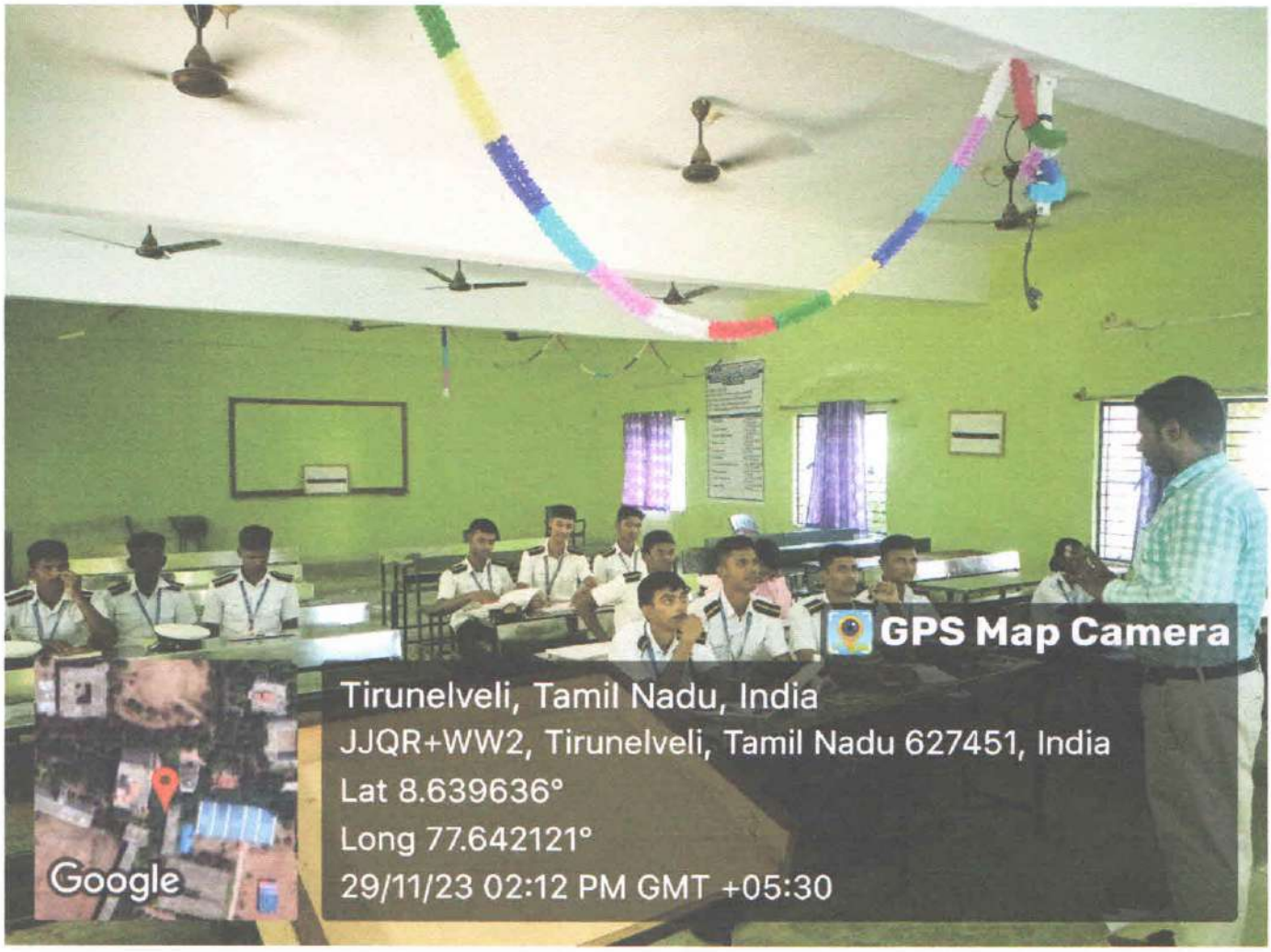
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DEPARTMENT OF MARINE ENGINEERING



Value Added Course on "VAC23003-Safety of Life at Sea (SOLAS)"
(27.11.2023 to 01.12.2023)

VAC Coordinator

Head of the Department / MARINE

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DEPARTMENT OF MARINE ENGINEERING

Value Added Course on "VAC23003-Safety of Life at Sea (SOLAS)"
(27.11.2023 to 01.12.2023)

Name: N. Vishwanath
Reg. No: 952222208016
Department: Marine Engineering
Year of study: 2022-2026
Date: 1/12/2023

93/100

S. [signature]

Time: 15 mins **Marks:** 15

Sl.No.	Question
1.	What are the technical provisions of SOLAS convention regarding Fire? a) Identify sources of fire on ships. b) Ship construction. c) Fire protection, fire detection and fire extinction d) Competency of ship's crew.
2.	When was the current SOLAS adopted? a) 1971 b) 1972 c) 1973 d) 1974
3.	Which convention approves the cargo securing manual? a) SOLAS b) STCW c) MARPOL d) ILO
4.	Which of the following are referred in SOLAS? a) ISPS Code b) LSA Code c) ISM Code d) All of them
5.	SOLAS does not apply to some type of vessels. Name any two types: a) Bulk carriers b) Chemical tankers. c) War ships/Fishing vessels. d) Container vessels.
6.	Which chapter of SOLAS refers to the requirements of ISM Code? a) XI b) VI c) IX d) IV

7.	The SOLAS Convention is regarded as the most important of all international treaties. What does this convention concern about?. a) Safety of merchant shipping companies. b) Safety of all sea ports. c) Safety of merchant ships. d) International sea trade.
8.	SOLAS has a specific chapter for a cargo. What cargo is it? a) Steel b) Grain c) Iron ore d) Crude oil
9.	Name any two statutory certificates required as per SOLAS: a) Load line certificate b) Tonnage certificate c) Passenger Ship Safety Certificate / Cargo Ship Safety Construction Certificate d) International pollution prevention certificate.
10.	Which of the following is Not part of the SOLAS requirement? a) ISPS Code b) LSA Code c) ISM Code d) STCW Code
11.	The first version of SOLAS was adopted in 1914. What was the reason for this adoption? a) Severe pollution of international waters. b) Prevent irregularities in bunker supplies. c) Titanic disaster. d) To control salvage claims.
12.	According to the SOLAS regulations, lifeboat falls must be renewed at intervals of how many years? a) 2.5 b) 3 c) 4 d) 5
13.	Port state control have detained vessels based on SOLAS convention. Which area on the vessels have maximum detentions? a) Navigation equipment. b) Safety Radio. c) Firefighting Appliances d) Lifesaving appliances
14.	What are the technical provisions of SOLAS convention regarding ship construction? a) Allowable changes to construction of fishing vessels. b) Capability of shipyards. c) Structure, subdivision and stability, machinery and electrical installations d) Construction of high speed crafts.
15.	What are the technical provisions of SOLAS convention on carriage? a) Carriage of arms and ammunition. b) Carriage of passengers. c) Carriage of cargoes and oil fuels / Carriage of dangerous goods d) Carriage of livestock.

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DEPARTMENT OF MARINE ENGINEERING

Value Added Course on "VAC23003-Safety of Life at Sea (SOLAS)"
(27.11.2023 to 01.12.2023)

Name: Thraneshwaran. A
Reg. No: 952222080013
Department: Marine
Year of study: 2022 - 2026
Date: 1.12.2023

93/100

S. Bhuvan

Time: 15 mins **Marks:** 15

Sl.No.	Question
1.	What are the technical provisions of SOLAS convention regarding Fire? a) Identify sources of fire on ships. b) Ship construction. c) Fire protection, fire detection and fire extinction d) Competency of ship's crew.
2.	When was the current SOLAS adopted? a) 1971 b) 1972 c) 1973 d) 1974
3.	Which convention approves the cargo securing manual? a) SOLAS b) STCW c) MARPOL d) ILO
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DEPARTMENT OF MARINE ENGINEERING

Value Added Course Conducted Report

Course	Value Added Course
Course Code	VAC23003
Title of the Course	Safety of Life at Sea (SOLAS)
Dates & Duration	27.11.2023 to 01.12.2023 & 30 Hours
Name(s) of the Resource Persons	Er.Manivannan S, Marine Engineer Officer / Merchant Navy
Name of the Course Coordinator	Mr.Ramkumar S, AP/Marine
No. of Participants	18

Course Outcomes

Upon the completion of this course, the students will able to

- Understand SOLAS convention adopted in 1974.
- Understand the contents of SOLAS.

Remarks of the Course Coordinator about the Course

Department of Marine Engineering, PSN College of Engineering and Technology organized the value added course on **"Safety of Life at Sea (SOLAS)"** from 27.11.2023 to 01.12.2023 at Marine Class Room PSNCET. This value added course focused on Learning Safety of Seafarer's Life at Sea and to improve their skills on Safe Working Practice Onboard Ship. 18 students registered and participated in this value added course. The course was well accepted by the students which are noted from the feedback given by them. This course may be conducted every year.

PO Mapping:

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
		✓			✓	✓	✓	✓	✓		✓	✓	✓

VAC Coordinator **Head of the Department / MARINE** **PRINCIPAL**

PSN College of Engineering and Technology

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DEPARTMENT OF MARINE ENGINEERING

FEEDBACK FORM FOR ACADEMIC EVENTS

Type of Event: conference/ seminar/ symposium/ guest lecture/ FDP/ Others... value added course

Date and Venue of Event:...27.11.2023 to 01.12.2023.. seminar hall

Organized by:.....Department of marine engineering.....

Please indicate the extent to which you agree with the following statements regarding the events:

S.NO	Statements	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1	Course Content provided at this event is relevant and useful	✓				
2	Visual lectures were clear and easy to understand	✓				
3	Presentation were interesting		✓			
4	The audibility of the Presentation were good	✓				
5	The venue was suitable	✓				
6	The course material handout was adequate		✓			
7	Overall, the event was worthwhile	✓				
8	Refreshments were suitable	✓				

What are the two most useful things you got out of the event?

I learned more above safety of life at sea

How could the event be improved?

I learned about safe working practice on boat ship

Other comments

Name of the participant with address:

.....B. Aneesiya,.....

.....4p2,....Kannangudi,...Devakottai (P),, Sivagangai (D). 630303.

PRINCIPAL

B. Aneesiya

STUDENT SIGNATURE

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Melathediyoor, Palayamkottai, Tirunelveli – 627 152

DEPARTMENT OF MARINE ENGINEERING

FEEDBACK FORM FOR ACADEMIC EVENTS

Type of Event: conference/ seminar/ symposium/ guest lecture/ FDP/ Others..VALUE ADDED COURSE

Date and Venue of Event:..27/11/2023..-..01/12/2023..(SEMINAR HALL)

Organized by:..DEPARTMENT OF MARINE ENGINEERING....

Please indicate the extent to which you agree with the following statements regarding the events:

S.NO	Statements	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1	Course Content provided at this event is relevant and useful	✓				
2	Visual lectures were clear and easy to understand	✓				
3	Presentation were interesting		✓			
4	Presentations were interesting		✓			
5	The venue was suitable	✓				
6	The course material handout was adequate	✓				
7	Overall, the event was worthwhile	✓				
8	Refreshments were suitable	✓				

What are the two most useful things you got out of the event?

1. I learned more about Safety of Life at Sea
2. I learned about Safety working Practice on boardship

How could the event be improved?

Other comments

Name of the participant with address:

.....P. SONA SUNDARAM.....D/NO: 224, PERUMAL KOVIL STREET,.
.....VARPATTU (POST), PONNAMBALAPATHY (TK), PUDUKOTTAI....622403

PRINCIPAL

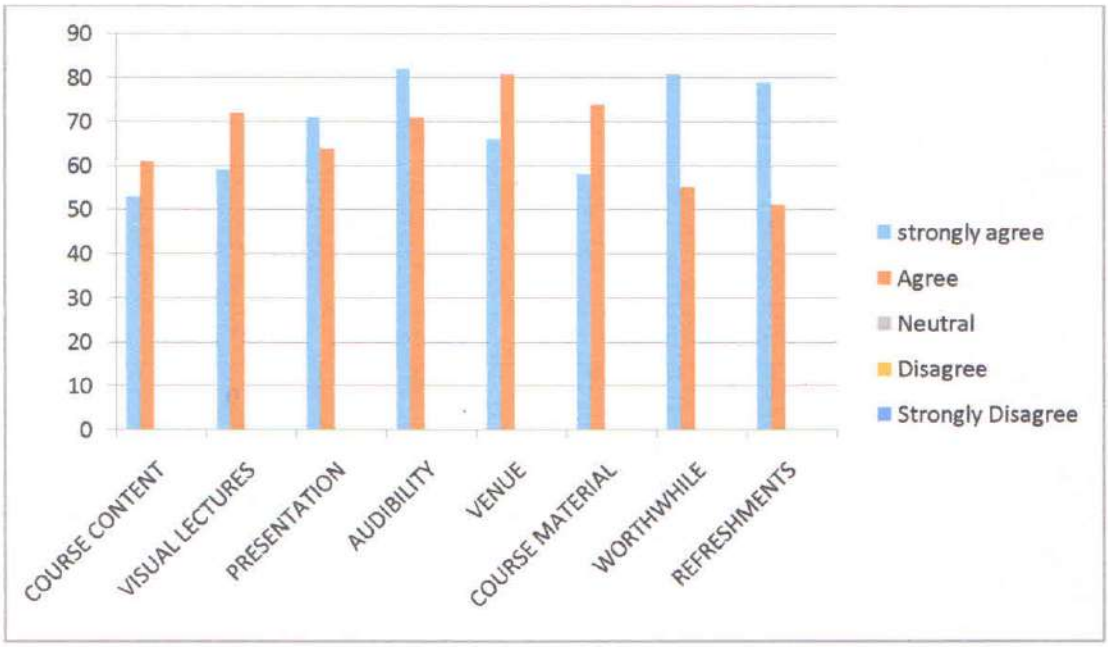
STUDENT SIGNATURE

PSN College of Engineering and Technology
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Melathediyoor, Palayamkottai, Tirunelveli – 627 152
DEPARTMENT OF MARINE ENGINEERING

Value Added Course on "VAC23003-Safety of Life at Sea (SOLAS)" (27.11.2023 to 01.12.2023)

FEEDBACK ANALYSIS

S.No	Code and Name of VAC Programme	Course Coordinator	Resource person	Duration (5 Days) (27/11/2023-01/12/2023)	Total hours
1	VAC23003-Safety of Life at Sea (SOLAS)	Mr.S.Ramkumar, Asst. Prof/Marine	Er.S.Manivannan, Marine Engineer Officer	10.00 am – 1.00 pm 2.00 pm – 4.00 pm	30



VAC CO-ORDINATOR

HoD/Marine

PRINCIPAL

PSN College of Engineering and Technology

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Melathediyoor, Palayamkottai, Tirunelveli – 627 152

DEPARTMENT OF MARINE ENGINEERING

(Approved by DG Shipping, Mumbai With A2 Grade)

CERTIFICATE OF PARTICIPATION

This is to certify that Mr/Ms .....ANANDHA NARAYANAN M.....
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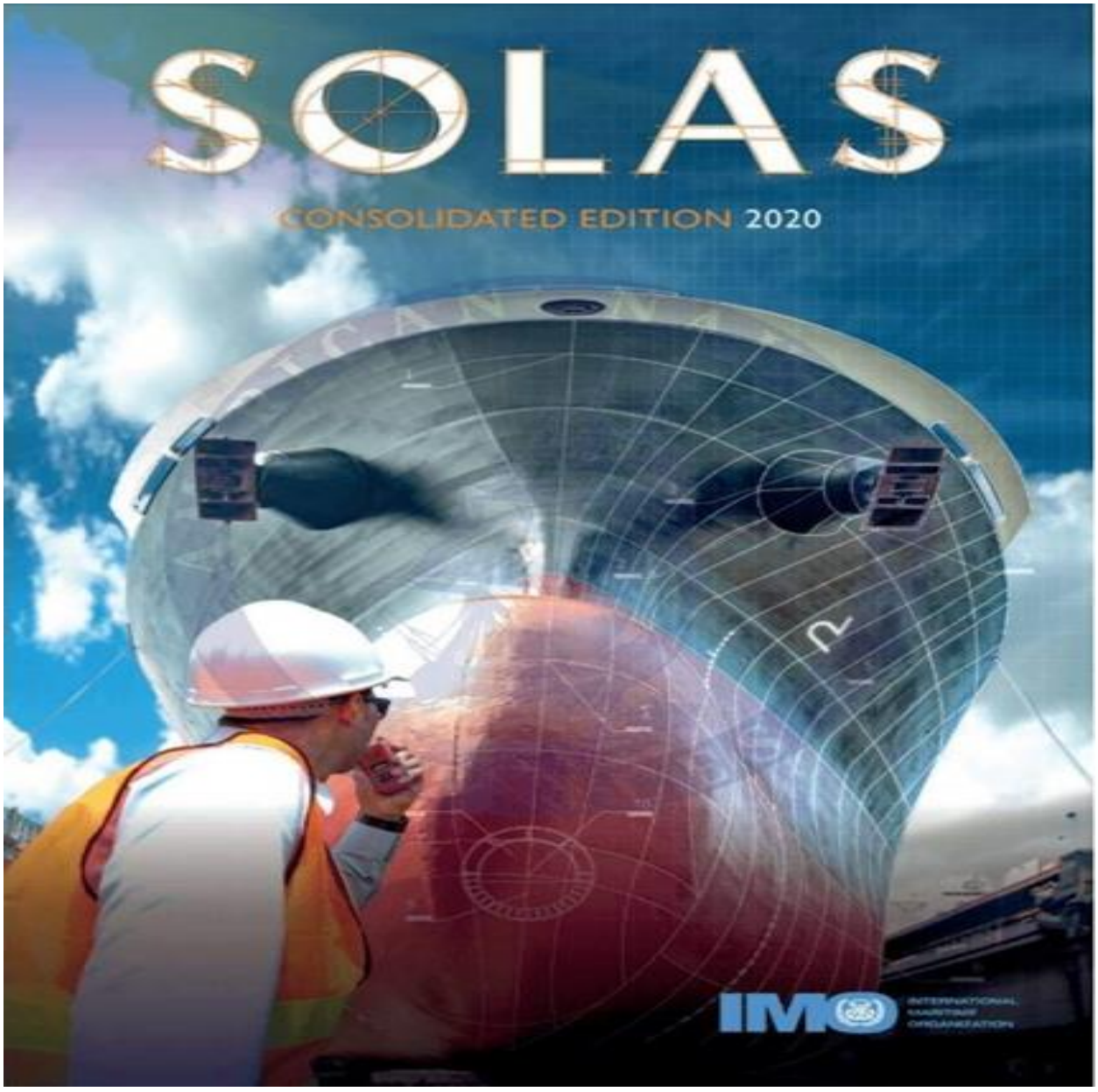
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DEPARTMENT OF MARINE ENGINEERING



Value Added Course

VAC23003-Safety of Life at Sea (SOLAS)

COURSE MATERIAL

Prepared by:
Er. S Manivannan,
Marine Engineer Officer
Indian Merchant Navy

Vision of the Institution:

- To emerge as a pioneer institute inculcating engineering education, skills, research, values and ethics

Mission of the Institution:

- To achieve greater heights of excellence in technical knowledge and skill development through innovative teaching and learning practices.
- To develop the state of art infrastructure to meet the demands of technological revolution.
- To improve and foster research in all dimensions for betterment of society.
- To develop individual competencies to enhance innovation, employability and entrepreneurship among students.
- To instill higher standards of discipline among students, inculcating ethical and moral values for societal harmony and peace.

Department vision

- To Produce competent, disciplined quality Engineers and administrators in accordance with global standards in Marine Engineering with capability of accepting new challenges.

Department mission

- To develop and train the students in order to meet dynamic needs and demands of the society.
- To promote research and continuing education.
- To enhance professional and entrepreneurial skills with ethical values through industry - institute interaction.

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1.0 International Treaties, Conventions, Protocols, Rules, Legislations and Regulations and Law of the Sea 1982 (UNCLOS)

1.1 Learning objectives: After reading this topic, you will be able to

- Understand international treaties, conventions and protocols.
- Understand law of the sea and zones of jurisdiction in sea.

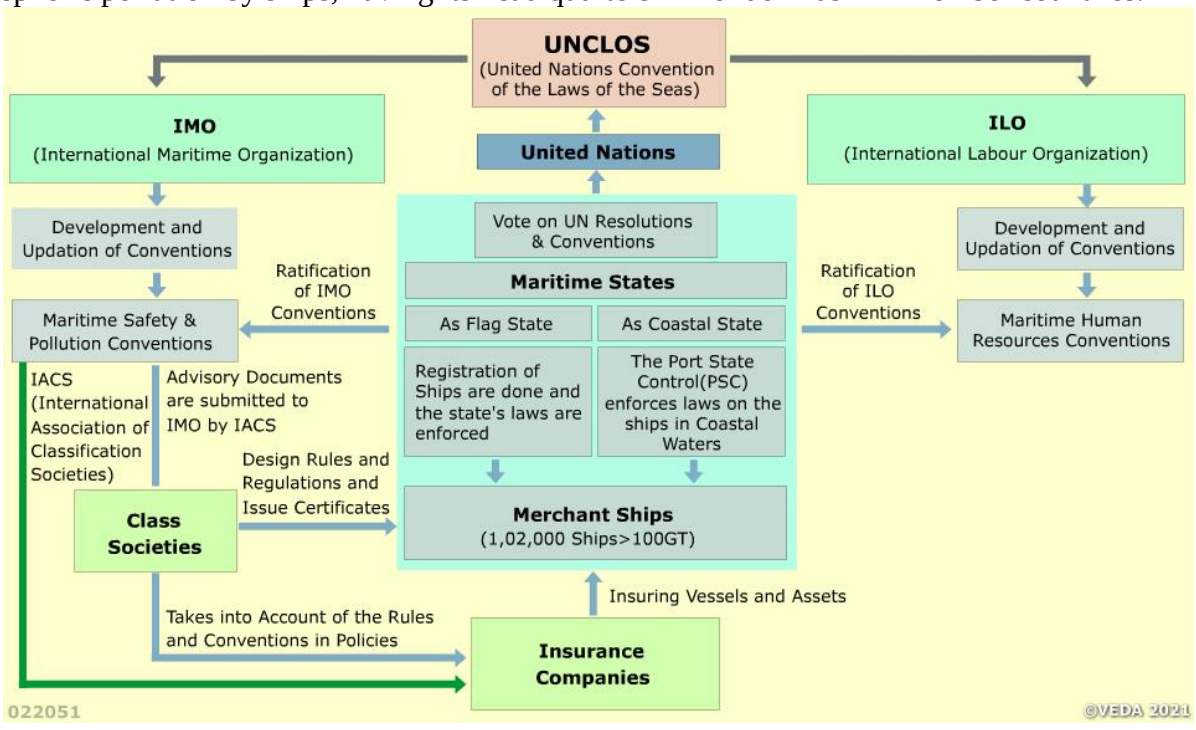
1.2 Introduction: This section provides detail of international conventions /protocols/ amendments /recommendations & guidelines/resolutions with respect to UNCLOS.

United Nations Organisation has been formed where in all the countries are members to manage the affairs, implement international agreements and manage them effectively. UN has got many organisations to manage, Trade, Health, Education etc.

"Treaty" means an international agreement concluded between States in written form and governed by international law, whether embodied in a single instrument or in two or more related instruments and whatever its particular designation;

A State may also sign a treaty "subject to ratification, acceptance or approval". In such a situation, signature does not signify the consent of a State to be bound by the treaty, although it does oblige the State to refrain from acts which would defeat the object and purpose of the treaty until such time as it has

To manage the maritime activities around the globe, International Maritime Organisation (IMO) has been formed. IMO – the International Maritime Organization – is the United Nations specialized agency with responsibility for the safety and security of shipping and the prevention of marine and atmospheric pollution by ships, having its head quarters in London has 174 member countries.



Some understanding of the following important instruments used by UN will be useful:

- Conventions
- Protocols
- Amendments
- Recommendations and guidelines
- Resolutions

1.3 Conventions: the other name for the Treaty is a written agreement between UN and the other countries. The convention process normally starts after an incident happens. The convention in a draft form can be proposed and introduced by a Member state or by group of member states. The UN Secretariat will study the feasibility of proposal, circulate among the members and put on vote on a notified day. If successful in voting, the convention is adopted. Please note that the convention is not a law in itself. A convention becomes a law when it is implemented in a country's own legal system meaning that only after every member country will have to make it as a law in their country, which takes a long time for the convention to come into effect.

1.4 Protocols: If there are some changes required to the Convention which has been adopted but not yet come into force, it is normally done by introducing a Protocol indicating changes to conventions. It is normally happens when a country is trying make the convention as a law in their country, get into some legal issues and request for some changes in the Convention.

1.5 Amendments: If some or all clauses of the convention are modified by the introduction of Protocol is an Amendment or we can simply state that Amendments are the reforms that are needed in a convention backed by the protocols.

1.6 Recommendations and guidelines: is a document linked to a convention, which is yet to be accepted, hence not a legally accepted / binding one for all the member states. Normally the member country(s) brought in the 'Recommendations and guidelines' has /have made it as a law in their country(s)

1.7 Resolutions: Resolutions are the finalized documents on any convention based on the recommendation or amendment and are accepted or passed by the required number of member countries

2.0 International Maritime Organisation (IMO) 1948

Convention on the International Maritime Organization
Adoption: 6 March 1948
Entry in force: 17 March 1958



2.1 The Geneva conference 1948

The Geneva conference opened in February 1948 and on 6 March 1948 the Convention establishing the Inter-Governmental Maritime Consultative Organization (IMCO) was adopted. (The name was changed in 1982 to International Maritime Organization (IMO).)

2.2 Organs of the IMO

The Organization consists of an Assembly, a Council, five main Committees and seven of Sub-Committees support the work of the main technical committees.

2.3 Five main committees.

The main functions of assembly, council and IMO committees
MSC-Marine safety committee
MEPC-Marine environment protection committee.
LEG- Legal committee
FAL- Facilitation committee.
TCC- Technical committee.

2.4 Sub-Committees

The MSC and MEPC are assisted in their work by a number of sub-committees which are also open to all Member States:

- Sub-Committee on Human Element, Training and Watchkeeping (HTW);
- Sub-Committee on Implementation of IMO Instruments (III);

- Sub-Committee on Navigation, Communications and Search and Rescue (NCSR);
- Sub-Committee on Pollution Prevention and Response (PPR);
- Sub-Committee on Ship Design and Construction (SDC);
- Sub-Committee on Ship Systems and Equipment (SSE); and
- Sub-Committee on Carriage of Cargoes and Containers (CCC)

3.0 SOLAS Convention 1974

3.1 Learning objectives:
After reading this topic, you will be able to understand

- SOLAS convention adopted in 1974.

3.2 Introduction: This section provides below for SOLAS:

1. Technical provisions.
2. Statutory certificates.
3. PSC detentions related to SOLAS.

International Convention for the Safety of Life at Sea, 1974 (SOLAS 1974)
Adoption: 1 November 1974
Entry into force: 25 May 1980

The SOLAS Convention in its successive forms is generally regarded as the most important of all international treaties concerning the safety of merchant ships. The first version was adopted in 1914, in response to the Titanic disaster, the second in 1929, the third in 1948, and the fourth in 1960. The 1974 version includes the tacit acceptance procedure - which provides that an amendment shall enter into force on a specified date unless, before that date, objections to the amendment are received from an agreed number of Parties.

As a result the 1974 Convention has been updated and amended on numerous occasions. The Convention in force today is referred to as SOLAS, 1974, as amended.

3.3 Technical provisions

The main objective of the SOLAS Convention is to specify minimum standards for the construction, equipment and operation of ships, compatible with their safety. Flag States are responsible for ensuring that ships under their flag comply with its requirements, and a number of certificates are prescribed in the Convention as proof that this has been done. Control provisions also allow Contracting Governments to inspect ships of other Contracting States if there are clear grounds for believing that the ship and its equipment do not substantially comply with the requirements of the Convention - this procedure is known as port State control. The current SOLAS Convention includes Articles setting out general obligations, amendment procedure and so on, followed by an Annex divided into 14 Chapters.

SOLAS covers nearly all types of commercial ships engaged in international voyages and has been ratified by 163 countries representing 99.17% of the world's shipping tonnage. As of 2018. SOLAS has been amended 63 times through either resolutions or conferences.

Flag States are responsible for ensuring that ships under their flag comply with the Flag State legislation developed to meet the Convention’s requirements. Certificates are prescribed in the convention as proof that this has been done. The SOLAS Convention also acts as the control instrument for other major conventions through control of certificates.

3.3.1 It contains technical provisions on:

- Construction – structure, subdivision and stability, machinery and electrical installations

- Fire protection, fire detection and fire extinction
- Life-saving appliances and arrangements
- Radio communications
- Safety of navigation
- Carriage of cargoes and oil fuels
- Carriage of dangerous goods
- Nuclear ships
- Management of the safe operation of ships
- Safety measures for high-speed craft
- Special measures to enhance maritime safety
- Special measures to enhance maritime security
- Additional safety measures for bulk carriers
- Verification of Compliance
- Safety Measures for Ships Operating in Polar Waters

The Port State Control provision in this Convention authorizes Member States to inspect vessels of other Member States if the Port State believes a vessel does not meet the requirements of the convention.

The Harmonized System of Survey and Certificates (HSSC) has been adopted in order to harmonize the survey and certification requirements of SOLAS, MARPOL and Load Line.

The 1974 Convention has been amended many times to keep it up to date.

3.5 Related Statutory Certificates

- Passenger Ship Safety Certificate
- Cargo Ship Safety Construction Certificate
- Cargo Ship Safety Equipment Certificate
- Cargo Ship Safety Radio Certificate
- Cargo Ship Safety Certificate
- Exemption Certificate
- Nuclear Passenger Ship Safety Certificate
- Nuclear Cargo Ship Safety Certificate

Detentions of ships by Port State Control under SOLAS related areas are many.

3.5.1 Some examples are

- FFA - 14%
- LSA - 12%
- General safety - 12%
- Safety navigation - 9%
- Safety radio - 3%
- Certificates & Log books - 3%
- ISM related matters - 3%
- Almost half of the ships detained are based on failure to conform to SOLAS convention.

3.6 SOLAS does not apply to

- Ships of war and troopships
- Cargo ships of less than 500 gross tonnage
- Ships not propelled by mechanical means
- Wooden ships of primitive build
- Pleasure yachts not engaged in trade
- Fishing vessels

4.0 Contents of SOLAS

4.1 Learning objectives:

After reading this topic, you will be able to understand

- The contents of SOLAS.

4.2 Introduction:

This section provides the contents of SOLAS chapters 1 to 14.

4.3 SOLAS PART 1:

Gives background about the Convention, definitions, certificates, Protocol of 1988. It consists of 14 Chapters and an Appendix showing the format of certificates

4.4 SOLAS Part 2:

Contains the Annexes
Annex 1 - Certificates and documents required to be carried on board ships
Annex 2 - List of Resolutions adopted by the SOLAS Conferences

4.5 SOLAS- Chapter I General provisions: Includes regulations concerning the survey of the various types of ships and the issuing of documents signifying that the ship meets the requirements of the Convention. The Chapter also includes provisions for the control of ships in ports of other Contracting Governments.

4.6 SOLAS- Chapter II-1 Construction – Structure, subdivision and stability, machinery and electrical installations: Includes regulations concerning

- Structure of ships
- Stability
- Subdivision, watertight and weathertight integrity
- Subdivision load line assignment for passenger ships
- Stability management
- Machinery installations
- Electrical installations
- Additional requirements for periodically unattended machinery spaces
- Alternative design and arrangements
- Ships using low – flashpoint fuels

The subdivision of passenger ships into watertight compartments must be such that after assumed damage to the ship's hull the vessel will remain afloat and stable. Requirements for watertight integrity and bilge pumping arrangements for passenger ships are also laid down as well as stability requirements for both passenger and cargo ships.

The degree of subdivision - measured by the maximum permissible distance between two adjacent bulkheads - varies with ship's length and the service in which it is engaged. The highest degree of subdivision applies to passenger ships.

Requirements covering machinery and electrical installations are designed to ensure that services which are essential for the safety of the ship, passengers and crew are maintained under various emergency conditions.

"Goal-based standards" for oil tankers and bulk carriers were adopted in 2010, requiring new ships to be designed and constructed for a specified design life and to be safe and environmentally friendly, in intact and specified damage conditions, throughout their life. Under the regulation, ships should have adequate strength, integrity and stability to minimize the risk of loss of the ship or pollution to the

marine environment due to structural failure, including collapse, resulting in flooding or loss of watertight integrity.

IGF Code – International code of safety for ships using gases or other low-flashpoint fuels was adopted in order to provide an International standard for the safety of ships using low-flashpoint fuel, other than ships covered by the IGC code.

IS Code – International code on Intact Stability is to provide mandatory requirements and recommended provisions relating to Intact Stability, based primarily on existing IMO Instruments.

Code on Noise levels on board ships is to provide measures for the protection against noise in machinery spaces of ships.

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4.7 SOLAS- Chapter II-2 Fire protection, fire detection and fire extinction: Includes detailed fire safety provisions for all ships and specific measures for passenger ships, cargo ships and tankers.

They include the following principles:

- Division of the ship into main and vertical zones by thermal and structural boundaries
- Separation of accommodation spaces from the remainder of the ship by thermal and structural boundaries
- Restricted use of combustible materials
- Detection of any fire in the zone of origin
- Containment and extinction of any fire in the space of origin
- Protection of the means of escape or of access for firefighting purposes
- Ready availability of fire extinguishing appliances
- Minimization of the possibility of ignition of flammable cargo vapor

FSS Code – International Code for Fire Safety Systems provides international standards of specific engineering specifications for fire safety systems required under this chapter.

FTP Code – International code for application of Fire Test Procedures is for use by the Administration and the competent authority of the flag state when approving products for installation in ships flying the flag of the flag state in accordance with the fire safety requirements.

4.8 SOLAS- Chapter III Life-saving appliances and arrangements: Includes requirements for life-saving appliances and arrangements, including requirements for life boats, rescue boats and life jackets according to type of ship.

4.9 SOLAS- Chapter IV Radio communications

The Chapter incorporates the Global Maritime Distress and Safety System (GMDSS)

All passenger ships and all cargo ships of 300 gross tonnage and upwards on international voyages are required to carry equipment designed to improve the chances of rescue following an accident, including satellite emergency position indicating radio beacons (EPIRBs) and search and rescue radar transponders (SARTs) for the location of the ship or survival craft.

4.11 SOLAS- Chapter V Safety of navigation: Chapter V identifies certain navigation safety services which should be provided by Contracting Governments and sets forth provisions of an operational nature applicable in general to all ships on all voyages.

This is in contrast to the Convention as a whole, which only applies to certain classes of ship engaged on international voyages. The subjects covered include the maintenance of meteorological services for ships; the ice patrol service; routing of ships; and the maintenance of search and rescue services.

This Chapter also includes a general obligation for masters to proceed to the assistance of those in distress (Good Samaritan Law-Law of the Sea) and for Contracting Governments to ensure that all ships shall be sufficiently and efficiently manned from a safety point of view.

The chapter makes mandatory the carriage of voyage data recorders (VDRs) and automatic identification system (AIS) for certain ships

4.12 SOLAS- Chapter VI Carriage of Cargoes: This Chapter applies to the carriage of all types of cargo (except liquids in bulk and gases in bulk) which, owing to their particular hazards to ships or persons on board, may require special precautions.
The IMSBC Code – International Maritime Solid Bulk Cargoes Code is to facilitate the safe stowage and shipment of solid bulk cargoes.

The CSS Code – Code of Safe practice for cargo stowage and securing applies to cargoes carried on board ships, other than solid and liquid bulk cargoes and timber stowed on deck.

The International Grain Code – International code for the safe carriage of grain in bulk, applies to engage in the carriage of grain in bulk.

The BLU Code – Code of practice for the safe loading and unloading of bulk carriers has been developed by IMO to minimize losses of bulk carriers.

The TDC Code - Code of safe practice for ships carrying Timber Deck Cargoes is to define structural requirements to ensure safety in the handling, stacking and transporting of containers

4.13 SOLAS- Chapter VII Carriage of dangerous goods

4.13.1 Part A - Carriage of dangerous goods in packaged form - includes provisions for the classification, packing, marking, labeling and placarding, documentation and stowage of dangerous goods. Contracting Governments are required to issue instructions at the national level and the Chapter makes mandatory the International Maritime Dangerous Goods (IMDG) Code, developed by IMO, which is constantly updated to accommodate new dangerous goods and to supplement or revise existing provisions.

The IMDG Code was first adopted by IMO in 1965 and has been kept up to date by regular amendments, including those needed to keep it in line with United Nations Recommendations on the Transport of Dangerous Goods which sets the basic requirements for all the transport modes.

4.13.2 Part A-1 - Carriage of dangerous goods in solid form in bulk - covers the documentation, stowage and segregation requirements for these goods and requires reporting of incidents involving such goods

4.13.3 Part B covers Construction and equipment of ships carrying dangerous liquid chemicals in bulk and requires chemical tankers built after 1 July 1986 to comply with the International Bulk Chemical Code (IBC Code)

4.13.4 Part C covers Construction and equipment of ships carrying liquefied gases in bulk and gas carriers constructed after 1 July 1986 to comply with the requirements of the International Gas Carrier Code (IGC Code)

4.13.5 Part D includes special requirements for the carriage of packaged irradiated nuclear fuel, plutonium and high-level radioactive wastes on board ships and requires ships carrying such products

to comply with the International Code for the Safe Carriage of Packaged Irradiated Nuclear Fuel, Plutonium and High-Level Radioactive waste on Board Ships (INF Code)

The chapter requires carriage of dangerous goods to be in compliance with the relevant provisions of the International Maritime Dangerous Goods Code (IMDG Code)

4.14 SOLAS- Chapter VIII - Nuclear ships: Gives basic requirements for nuclear-powered ships and is particularly concerned with radiation hazards It refers to detailed and comprehensive Code of Safety for Nuclear Merchant Ships which was adopted by the IMO Assembly in 1981

4.15 SOLAS- Chapter IX - Management for the Safe Operation of Ships: The Chapter makes mandatory the International Safety Management (ISM) Code, which requires a safety management system to be established by the shipowner or any person who has assumed responsibility for the ship (the "Company")

4.16 SOLAS-Chapter X - Safety measures for high-speed craft: The Chapter makes mandatory the International Code of Safety for High-Speed Craft (HSC Code)

4.17 SOLAS-Chapter XI-1 - Special measures to enhance maritime safety: The Chapter clarifies requirements relating to authorization of recognized organizations (responsible for carrying out surveys and inspections) in accordance with RO Code – Code for recognized organization.

Bulk carriers and oil tankers as defined in other chapters shall be subject to an enhanced program of inspections in accordance with the ESP Code – International Code on the enhanced program of inspections during surveys of bulk carriers and oil tankers.

Casualty investigation code – Code of the international standards and recommended practices for a safety investigation into a marine casualty or marine incident is to provide a common approach for States to adopt in the conduct of marine safety investigations into marine casualties and marine incidents.

4.18 SOLAS-Chapter XI-2 - Special measures to enhance maritime security: The Chapter was adopted in December 2002 and entered into force on 1 July 2004.

ISPS Code – The International Ship and Port Facility Security Code is to establish an international framework to detect/assess security threats and take preventive measures against security incidents affecting ships or port facilities.
Regulation XI-2/2 of the chapter enshrines the International Ship and Port Facilities Security Code (ISPS Code). Part A of the Code is mandatory and part B contains guidance as to how best to comply with the mandatory requirements.

Regulation XI-2/8 confirms the role of the Master in exercising his professional judgment over decisions necessary to maintain the security of the ship. It says he shall not be constrained by the Company, the charterer or any other person in this respect.

Regulation XI-2/6 requires all ships to be provided with a ship security alert system.

Regulation XI-2/10 covers requirements for port facilities, providing among other things for Contracting Governments to ensure that port facility security assessments are carried out and that port facility security plans are developed, implemented and reviewed in accordance with the ISPS Code. Other regulations in this chapter cover the provision of information to IMO, the control of ships in

port, (including measures such as the delay, detention, restriction of operations including movement within the port, or expulsion of a ship from port), and the specific responsibility of Companies.

4.19 SOLAS-Chapter XII - Additional safety measures for bulk carriers: The Chapter includes structural requirements for bulk carriers over 150 meters in length.

4.20 SOLAS-Chapter XIII – Verification of Compliance: Under III code – IMO Instruments Implementation Code - contracting governments shall use the provisions of this code for implementation in the execution of their obligations and responsibilities in SOLAS Convention.

4.21 SOLAS-Chapter XIV - Safety Measures for Ships Operating in Polar Waters: The Polar Code - International code for Ships Operating in Polar Waters has been developed to increase the safety of ships operation and mitigate the impact on the people and environment in the remote, vulnerable and potentially harsh Polar Waters. The chapter was made mandatory, from 1 January 2017, the Introduction and part I-A of the International Code for Ships Operating in Polar Waters (the Polar Code).
Brief discussion on the reasons for Bulk Carrier losses in the early 1990s and how provisions detailed in Chapter XII of SOLAS 74 as amended will contribute towards the safety of bulk-carriers:

A study into bulk carrier survivability carried out by the International Association of Classification Societies (IACS), at the request of IMO, found that if a ship is flooded in the forward hold, the bulkhead between the two foremost holds may not be able to withstand the pressure that results from the sloshing mixture of cargo and water, especially if the ship is loaded in alternate holds with high-density cargoes (such as iron ore). If the bulkhead between one hold and the next collapses, progressive flooding could rapidly occur throughout the length of the ship and the vessel would sink in a matter of minutes.

The IACS concluded that the most vulnerable areas are the bulkhead between numbers one and two holds at the forward end of the vessel and the double bottom of the ship at this location. It proposed that particular attention should be paid to these areas during special surveys of ships and, where necessary, reinforcements should be carried out.

A study by the U.S. Maritime Administration {MARAD) found that a typical midsize bulk carrier should survive all one-hold flooding so long as the ship is not suffering from metal wastage and undetected cracks, but flooding of any two holds would have disastrous consequences.

Recommendations relating to the design and construction of bulk carriers' issues considered by IMO included:

1. Strength of hatch covers and coamings;
2. Freeboard and bow height;
3. Reserve buoyancy at fore end, including forecastles;
4. Structural means to reduce loads on hatch covers and forward structure; and
5. Foredeck and fore end access

The highest loss of life has been attributed to Cargo failure (liquefaction), totaling 101 lives lost or 53.7% of total loss of life resulted from the 9 casualties.

The most common reported cause of ship losses has been Grounding, with 19 losses or 39.6 % of total losses.

Losses due to Flooding for 6 ships (12.5%) and have been significant.

Losses of lives resulting from 6 ships lost with Unknown causes accounted for 61 lives, or 32.4% of total life losses; among these 6 ship losses were the 2 casualties in 2017 resulting in 32 lives lost.

4.21.1 Causes of Ship's hull damage:

- Damage to side shell, externally through contact with docksides or tugs and, internally from impact by cargo dislodging equipment during discharge, can result in initiating fractures and/or fatigue of the structure. In single side-skin bulk carriers, bulkheads, trunks and ballast tank boundaries, can present "hard spots" that concentrate forces where the change in construction occurs (e.g. longitudinal to transverse framing). This may lead to undetected fractures.
- Internal degradation through corrosion may be accelerated through chemical action from certain cargoes. Welds in particular may be subject to "grooving" corrosion, in which the material forming the weld corrodes at a faster rate than the plating to which it is attached. Fatigue failure may result due to loss of cross-sectional area in the plating joints.
- Damage to bow plating such as is possible through impacts associated with swinging or loosely stowed anchors may cause an initiating fracture or fatigue in bow shell plating that could lead to failure and subsequent flooding. Internal integrity of forward spaces (that are usually used for ballast and/or stores) is therefore of vital importance. Corrosion degradation will seriously reduce the ability of plating and stiffening to withstand the forces to which it will be subjected. In larger ships, partially filled forepeak tanks may set up destructive sloshing forces unless the tank structure is designed for this.
- External forces horizontal and/or vertical - may cause hatch cover dislodgement. The cargo hatchway, if it loses its protection in this way, is a major access for water ingress and a serious threat to the integrity of the hull.
- When a vessel requires repairs to damaged equipment or to the hull it is necessary for the work to be carried out to the satisfaction of the classification society surveyors. In order that the ship maintains its class, approval of the repairs undertaken must be obtained from the surveyors either at the time of the repair or at the earliest opportunity.

4.22 Provisions as detailed in Chapter XII of SOLAS 74 as amended, has contributed towards the safety of bulk-carrier as follows:

The regulations state that all new bulk carriers of 150 metres or more in length (built after 1 July 1999) carrying cargoes with a density of 1000 kg/m3 and above should have sufficient strength to withstand flooding of any one cargo hold, taking into account dynamic effects resulting from the presence of water in the hold and taking into account recommendations adopted by IMO.

For existing ships (built before 1 July 1999) carrying bulk cargoes with a density of 1780 kg/m^3 and above, the transverse watertight bulkhead between the two foremost cargo holds and the double bottom of the foremost cargo hold should have sufficient strength to withstand flooding and the related dynamic effects in the foremost cargo hold.

Cargoes with a density of 1780 kg/m3 and above include iron ore, pig iron, steel, bauxite and cement. Less dense cargoes, but with a density of more than 1000 kg/m3, include grains such as wheat and rice, and timber.

Chapter XII allows surveyors to take into account restrictions on the cargo carried when considering the need for, and the extent of, strengthening of the transverse watertight bulkhead or double bottom. When restrictions on cargoes are imposed, the bulk carrier should be permanently marked with a solid triangle on its side shell.

The date of application of chapter XII to existing bulk carriers depends on their age. Bulk carriers

which are 20 years old and over on 1 July 1999 were to comply by the date of the first intermediate or periodical survey after that date, whichever was sooner. Bulk carriers aged 15±20 years had to comply by the first periodical survey after 1 July 1999, but not later than 1 July 2002. Bulk carriers less than 15 years had to comply by the date of the first periodical survey after the ship reaches 15 years of age, but not later than the date on which the ship reached 17 years of age.

In order to safeguard from these factors, bulk carrier ships are now added to the enhanced survey program (ESP) along with oil tanker ships. Some important points that help in improving the overall safety of the ship:

4.22.1. Reinforcing the corrugated transverse bulkhead

Corrugated bulkheads can be seen as a transverse arrangement on the inside of the cargo holds. The concave shape is achieved by the use of plate rolls or heat line bending. The corrugated has a higher strength than a flat plate when subjected to bending moments during cargo operations or even at sea.

Reinforcing the corrugated transverse bulkhead installed between No.1 and No.2 cargo holds and the double bottom of No. 1 hold. This ensures watertight integrity to be maintained at all-time even when side hold or bottom is damaged.

4.22.2. Water Ingress Alarm

- Water level detection and alarm system is to be located on the navigation bridge and be capable of detecting water ingress at all cargo holds and spaces and ballast tank forward of collision bulk head. In general, FPT, Bosn's Store, F'cle Space excluding chain lockers are considered as these spaces.
- Water levels are to be detected at as close to the center line (within B/6m from center line), or at both the port and starboard sides of the cargo hold Ship Bilge alarms and water ingress detectors had already been provided in accordance with SOLAS Regulation 9 Chapter XII.
- The water ingress alarm is in place seeing as the ingress of water may taint or corrupt the cargo/commodity being carried on a bulk carrier.
- In each cargo hold, the systems are to give alarms when the water level reaches the following (a) and (b) at the aft end of the cargo hold. (a) a height of 0.5m above the inner bottom. (b) a height not less than 15% of the depth of the cargo hold but not more than 2.0m.
- In any ballast tank forward of the collision bulk head, the system is to give an alarm when the liquid in the tank reaches a level not exceeding 10% of the tank capacity. In any dry or void space other than chain locker, any part of which extends forward of the foremost cargo hold and the volume of which exceeds 0.1% of the ship's maximum displacement volume, the system is to give an alarm at a water level of 0.1m above the deck.
- In any dry or void space other than chain locker, any part of which extends forward of the foremost cargo hold and the volume of which exceeds 0.1% of the ship's maximum displacement volume, the system is to give an alarm at a water level of 0.1m above the deck.

4.22.3. Avoiding Deck Wetness

Increasing the Integrity of fore-deck fittings on bulk carriers to avoid the problem of deck wetness i.e. flowing water on ship's deck due to rough weather, also known as green water loading

4.22.4. Fitting Bulwark

Bulwark is the extension of a ship's sides above the level of the deck. Not only does it fulfil the purpose of strengthening but it also prevents the outflow of water or, God forbid, oil! Fitting of suitable bulwark in the fore part of the ship as per the requirement to avoid green water condition

4.22.5. Reinforcing Hatch Cover
Hatch covers are the primary arrangement for protecting the cargo being carried onboard; also, to maintain the watertight integrity as well as to protect the cargo from the sea, sun and the weather in general. Reinforce the hatch cover which is located at the forward of the ship within 0.25L, where L is length of the ship

4.22.6. Maintain Water Tightness
Maintaining water tightness of all the hatch covers to avoid water ingress. A fire hose may be used on high pressure to subject the hinges to a water jet at full force that can be used to determine the watertight integrity of the hatch.

4.22.7. Double Skin Side Shell Frame
Side shell frame to be double skin ensuring extra safety in case of structural damage. In layman terms, if one layer is compromised or breached, there is another layer to ensure that the hull is not entirely breached which might result in a lot of damage to the ship and its contents

4.22.8. Cargo Hatch Cover Alarm
Installation of an alarm system for opening and closing of hatch cover for additional safety. When there is sudden downpour in port it is essential to close the hatch swiftly to ensure that the cargo does not get tainted due to the water. The presence of an alarm system is essential for the safety of crew as well as the cargo

4.22.9. Anti-Corrosion Paint
A good anti corrosive coating on the ship's structure to avoid damage due to corrosion. The ship is subjected to great stresses from the sea which might result in causing the structure to reduce in strength.

International Maritime Solid Bulk Cargoes (IMSBC) Code – facilitates the safe stowage and shipment of solid bulk cargoes by providing information on the risks associated with their shipment, and the procedures to be adopted for carriage.

The primary aim of the International Maritime Solid Bulk Cargoes **Code (IMSBC Code)**, which replaces the **Code** of Safe Practice for Solid Bulk Cargoes (BC **Code**), is to facilitate the safe stowage and shipment of solid bulk cargoes by providing information on the dangers associated with the stowage and shipment of certain types of solid bulk cargoes and instructions on procedures to be adopted when shipment of such bulk cargoes is contemplated.

5.0 Ship Certificates / Documents under SOLAS

5.1 Learning objectives:
After reading this topic, you will be able to understand

- The ship certificates under SOLAS.

5.2 Introduction: This section details the required ship documents and certificates under SOLAS.

5.3 Intact stability booklet SOLAS 1974, regulations II-1/22 and II-1/25-8;1988 LL Protocol, regulation 10-Every passenger ship regardless of size and every cargo ship of 24 meters and over shall be inclined on completion and the elements of their stability determined.

5.4 Damage control plans and booklets: SOLAS 1974, regulations II-1/23, 23-1, 25-8; MSC/Circ.919 On passenger and cargo ships, there shall be permanently exhibited plans showing clearly for each deck and hold the boundaries of the watertight compartments, the openings therein

with the means of closure and position of any controls thereof, and the arrangements for the correction of any list due to flooding. Booklets containing the aforementioned information shall be made available to the officers of the ship

5.5 Minimum safe manning document: SOLAS 1974 (2000 amendments), regulation V/14.2. Every ship to which chapter I of the Convention applies shall be provided with an appropriate safe manning document or equivalent issued by the Administration as evidence of the minimum safe manning

5.6 Fire safety training manual: SOLAS 1974 (2000 amendments), regulation II-2/15.2.3. A training manual shall be written in the working language of the ship and shall be provided in each crew mess room and recreation room or in each crew cabin.

5.7 Fire Control plan/booklet SOLAS: 1974 (2000 amendments), regulation II-2/15.2.4. General arrangement plans shall be permanently exhibited for the guidance of the ship's officers, showing clearly for each deck the control stations, the various fire sections together with particulars of the fire detection and fire alarm systems and the fire extinguishing appliances etc.

On board training and drills record SOLAS 1974 (2000 amendments), regulation II-2/ 15.2.2.5
Fire drills shall be conducted and recorded in accordance with the provisions of regulations III/19.3 and III/19.5

5.8 Fire safety operational booklet (SOLAS Manual): The fire safety operational booklet shall contain the necessary information and instructions for the safe operation of the ship and cargo handling operations in relation to fire safety

5.9 Voyage data recorder system - certificate of compliance SOLAS 1974, regulation V/18.8: The voyage data recorder system, including all sensors, shall be subjected to an annual performance test. The test shall be conducted by an approved testing or servicing facility to verify the accuracy, duration and recoverability of the recorded data

5.10 Document of Compliance: A document of compliance shall be issued to every company which complies with the requirements of the ISM Code. A copy of the document shall be kept on board. SOLAS 1974, regulation IX/4; ISM Code, paragraph 13

5.11 Safety Management Certificate: A Safety Management Certificate shall be issued to every ship by the Administration or an organization recognized by the Administration.

5.12 International Ship Security Certificate: An International Ship Security Certificate (ISSC) shall be issued to every ship by the Administration or an organization recognized by it to verify that the ship complies with the maritime security provisions of SOLAS chapter XI-2 and part A of the ISPS Code

5.13 Ship Security Plan and associated records: Each ship shall carry on board a ship security plan approved by the Administration. The plan shall make provisions for the three security levels as defined in part A of the ISPS Code. Records of the following activities addressed in the ship security plan shall be kept on board for at least the minimum period specified by the Administration:

- Training, drills and exercises;
- Security threats and security incidents;
- Breaches of security;
- Changes in security level;

communications relating to the direct security of the ship such as specific threats to the ship or to port facilities the ship is, or has been, in;

- Internal audits and reviews of security activities;

- Periodic review of the ship security assessment;
- Periodic review of the ship security plan;
- Implementation of any amendments to the plan; and
- Maintenance, calibration and testing of any security equipment provided on board, including testing of the ship security alert system (SSAS)-SOLAS 1974 (2002 amendments), regulation XI-2/9; ISPS Code part A, sections 9 and 10

Statutory certification of ships is done on the basis of IMO instruments. The instruments on which this certification is based are:

- Load Line Convention 1966
- Tonnage Convention 1969
- SOLAS Convention 1974
- MARPOL Convention 1973/78
- ISM Code
- ISPS Code.

The above conventions and the codes relate to carriage of various types of cargo. They set uniform standards to facilitate acceptance of a ship registered in one country in the waters and ports of another, with the general aim to promote safety at sea and protection of the environment. These standards are commonly referred to as 'statutory' requirements, because they are made part of a country's statutory law, require ships to be constructed, equipped and maintained to certain norms which are to be ensured by the Flag State administrations.

The administrations may entrust the task wholly or partly to certifying authorities such as

- The Flag States themselves (Flag State)
- A Recognized Organization (RO)
- A Classification Society (Class).

When a vessel is found to conform to the standards set forth, the relevant statutory certificates are issued to the vessel by the certifying authority with respect to the type of vessel.

The RO and Class are responsible and accountable to the Flag State for the work that they carry out on their behalf. The principles of inspection and survey verify that a ship is in compliance with applicable requirements at the time of the survey. The scope of these inspections and surveys regarding safety and pollution prevention are laid down by the Flag State in compliance with the relevant international conventions to which it is a signatory.

6.0 Regulatory Surveys and Penalities

6.1 Learning objectives:

After reading this topic, you will be able to understand

- Regulatory surveys and penalties for noncompliance.

6.2 Introduction: This section provides the basic details of:

1. Regulatory certificates and penalties.
2. Issue and control of statutory certificates.
3. Validity of certificates.
4. Liability for prosecution.
5. Fees and expenses.

- IMO requires that Flag state enforce procedures for certification and survey of vessels registered under their flags.
- Regulation set penelties for non compliance with certificate regulation.

Survey requirements and penalty for non-compliance

Certificates and surveys

Passenger certificate

Passenger ship safety certificate

Cargo ship safety construction

Cargo ship safety equipment

Radio certificate

Load line certificate

Certificate of registry

IOPP

Passenger certificate

Survey

The Passenger Certificate stipulates how many passengers can be carried, where they must be carried and the vessel's trading limits.

This certificate is valid for 12 months.

Penalty

If a vessel carries more passengers than she is certified to carry, the owner and Master are each guilty of an offence and liable, on summary conviction, to a maximum fine of £ 50,000 or, on conviction on indictment, to a fine.

If a ship proceeds to sea without being surveyed and without a certificate, the Master and Owner are each guilty of an offence and liable, on summary conviction, to a maximum fine of £ 1,000 or, on conviction on indictment, to a fine.

Passenger ship safety certificate

Survey

The Passenger Ship Safety Certificate is a general safety certificate covering structure, watertight subdivisions, subdivision load lines, damage control, bilge pumping arrangements, electrical equipment and installations, boilers and machinery, including steering gear, fire detection and protection, means of escape, anchors and cables, ropes, safety equipment and radio installation.

This certificate is valid for 12 months.

Penalty

If a ship proceeds or attempts to proceed on a voyage without being surveyed, or without a certificate, the Owner and Master are each guilty of an offence and liable, on summary conviction, to a maximum fine of £ 1,000 or, on conviction on indictment, up to 2 years imprisonment, or a fine, or both, and the ship detained.

Cargo ship safety construction certificate

Survey

Surveys cover structural strength, watertight divisions, watertight doors, ballast and bilge pumping and drainage arrangements, machinery, boilers and other pressure vessels, fuel and lubricating systems, steering gear, electrical installations, anchors and cables, with additional coverage for some tankers to include precautions against oil spillage, and ventilation.

Penalty

If a ship proceeds, or attempts to proceed to sea without being surveyed, or without to sea without being surveyed, or without a certificate, the Owner and Master are each guilty of an offence, and liable, on summary conviction, to a maximum fine of £ 2,000 or, on conviction on indictment, to imprisonment for up to 2 years, or a fine, or both, and the ship may be detained.

Cargo ship safety equipment certificate

Survey

The survey covers life-saving appliances, fire protection, and navigational equipment. Certificates cover aspects of safety on the vessel, including construction, equipment, load lines and manning. In the U.K. the initial survey is to be carried out by an MSA surveyor.

The MSA require that at least one subsequent survey in a period of 5 years is to be carried out by one of their surveyors. At any other time surveys may be carried out by a nominated surveyor.

Penalty

If a ship proceeds, or attempts to proceed to sea without being surveyed, or without a certificate, the Owner and Master are each guilty of an offence, and liable, on summary conviction, to a maximum fine of £ 2,000 or, on conviction on indictment, to imprisonment for up to 2 years, or a fine, or both, and the ship may be detained.

Radio certificate

Survey

The certificate contains information related to the number of operators which are to be carried, the hours of listening watches and whether the auto-alarm is fitted.

Penalty

If a ship goes, or attempts to go to sea without complying with the regulations, the Owner or Master are each guilty of an offence and liable, on summary conviction, to a maximum fine of £ 1,000 or, on conviction on indictment, to not more than 2 years imprisonment, or a fine, or both, and the ship detained.

Load line certificate

Survey

The Owner of a ship applies to an Assigning Authority for the assignment of freeboards. The Assigning authority decides which load lines are to be marked on the ship's side and their position, together with the position of the deck line and load line mark. The rules describe these marks as "appropriate marks". A vessel with valid certificates is not necessarily 'safe'. It is the actual condition of the ship and its equipment which will decide whether or not the ship is 'safe'. A ship is to be loaded so that, in salt water, or on entry into salt water, with no list, the appropriate load line is not submerged.

Penalty

The overloading of a ship is an offence for which the Owner or Master are liable, on conviction on indictment, to a fine or, on summary conviction to a maximum fine of £ 1,000, and an additional fine, which the court thinks fit, reflecting the increased earning capacity brought about by the offence.

Certificate of registry

Survey

This establishes the ship's nationality, measurements and tonnage for identification purposes. It is not a document of title, and is not to be taken as legal evidence of registered ownership.

It is one of the most important documents on the ship. It should always accompany the ship and remain in the master's custody. It is valid for a maximum of 5 years; if ownership of the ship changes in the 5-year period, a new 5-year certificate will be issued.

Penalty

It is an offence on the part of an Owner or Master if any mark on a ship is defaced, altered, allowed to become illegible, covered or concealed. Such an offence is punishable, on conviction on indictment, with a fine, or, on summary conviction, with a maximum fine of £ 2,000.

IOPP

Survey

International Oil Pollution Prevention Certificate:
Oil tankers of 150 GT and over, and other ships of 400 GT and over, are to be surveyed before being put into service or before the first issue of an International Oil Pollution Prevention Certificate or United Kingdom Oil Pollution Prevention (UKOPP) Certificate.

The Certificate is valid for 5 years, subject to annual and intermediate surveys.

Penalty

If there is any contravention of these regulations the owner and master of the ship shall each be guilty of an offence punishable on summary conviction by a fine not exceeding the statutory maximum, or on conviction on indictment, by a fine.

6.3 Issue and Control of Statutory Certificates

6.3.1 Issue of Statutory Certificates: The owner, who intends to obtain statutory certificates for his ship is first required to obtain the ship's Certificate of Registry from the Flag State Administration. Having done so, the owner is to submit a written application to the certifying authority for the issue of the statutory certificates. This may be the Flag Administration, or a Classification Society/Recognized Organization which acts on behalf of the Flag State Administration.

6.4 Interval between Surveys: Surveys are to be carried out at prescribed intervals:

- These surveys are known as yearly, intermediate (2.5 years), and renewal or special (5 years)
- There is a time "window" of 6 months (3 months before to 3 months after) within the anniversary date of the certificate for the annual, intermediate and periodical surveys
- The renewal or special survey may be completed within 3 months before the expiry date of the existing certificate.

6.5 Rewriting a Statutory Certificate: When the descriptions on a statutory certificate are to be changed, the owner or Master of a ship is to present, at the earliest opportunity, to the Flag State Administration (or the Classification Society/Recognized Organization acting on behalf of the Flag State Administration), an application, along with the ship's Certificate of Registry (issued by the Flag State Administration), for rewriting the relevant certificate.

6.6 Reissue of a Statutory Certificate: When a statutory certificate is soiled, or lost, the owner or Master is to submit a written application to the Flag Administration (or the Classification Society/Recognized Organization acting on behalf of the Flag State Administration) for re-issue of the certificate.

6.7 Endorsement of a Statutory Certificate: The owner or Master of a Ship, who wishes to obtain an endorsement on a statutory certificate, is to submit a written application to the Flag State Administration (or the Classification Society/Recognized Organization acting on behalf of the Flag State Administration), along with the statutory certificate on which the endorsement is required.

6.8 Returning a Statutory Certificate: The owner or Master of a ship shall return a statutory certificate to the Flag State Administration (or the Classification Society/Recognized Organization acting on behalf of the Flag State Administration) under the following circumstances:

- The ship sinks or is scrapped
- The Classification of the ship is deleted
- The validity of the statutory certificate is expired
- The descriptions on the statutory certificate are changed, and a revised certificate is issued
- When the original certificate is discovered after a reissue has been obtained. (The discovered certificate is to be returned.)
- When asked by the Flag State Administration (or the Classification Society/Recognized Organization acting on behalf of the Flag State Administration), to return the statutory certificate.

6.9 Validity of Certificates: The Flag State Administration makes the rules governing validity of a ship's statutory certificates:

- There is standard validity of 5 years for all cargo ships' certificates and 12 months for the Passenger Ship Safety Certificate
- Extension of certificates is possible, limited to 3 months, to enable a ship to complete its voyage, or 1 month, for ships engaged in short voyages
- When an extension has been granted, the period of validity of the new certificate starts from the date of expiry of the existing certificate before its extension
- Where a renewal survey has been completed and a certificate cannot be issued or placed aboard the ship before the expiry date of the existing certificate, the existing certificate may be

endorsed and will be accepted as the new certificate for a period of up to 5 months from the expiry date.

6.10 Cessation of Validity: A certificate shall cease to be valid:

- If its period of validity has been exceeded and the certificate has not been extended as permitted
- If Annual and Intermediate Surveys have not been carried out in accordance with Regulations, and the Certificate has not been endorsed
- Upon the transfer of a ship to the flag of another government.

6.11 Cancellation of a certificate: A certificate may be cancelled if:

- The certificate was issued on false or erroneous information
- Since the last survey as required by regulations, the structure/equipment/machinery has sustained damage, or is otherwise deficient
- The structural strength of the ship is lowered to unsafe standards
- A new certificate is issued
- The ship is no longer registered with the relevant Flag state.

6.12 Liability for Prosecution: The following offenses invite liability for prosecution:

- Intentionally altering a certificate issued to a vessel
- Intentionally making a false certificate
- During any survey knowingly or recklessly furnishing false information
- Using, lending or allowing to be used by another, with intent to deceive, a certificate issued to a vessel
- Failure to surrender a certificate when so required.

6.13 Fees and Expenses: Fees and expenses are charged in the following cases:

- A statutory certificate is issued
- A statutory certificate is reissued or rewritten
- Endorsement is made for a statutory certificate
- The validity of a statutory certificate is extended
- When inspection by the society is requested by the administration of the Port State
- When the survey for maintenance of the validity of the statutory certificate is carried out.

7.0 Documents and Certificates to be carried on board

7.1 Learning objectives:

After reading this topic, you will be able to list out

- Required documents and certificates to be carried on board.

7.2 Introduction:

This section provides the lists as below:

1. Documents to be carried on board.
2. Certificates to be carried on board.
3. ILO requirements of documents and certificates.
4. Additional statutory certificates required for Indian flag vessels.

7.3 Documents to be carried on board

- General arrangement plan
- Fire and safety plan
- Plan of piping arrangement - bilge line arrangement, oily water separator, ballast and fuel tank arrangement, including capacity, fresh water and sanitary arrangements
- Plan of electrical arrangement
- Plan of crew accommodation
- Navigation light arrangement
- Details of remote machinery stop valves and vent flaps
- Tank capacity, line diagrams, fire control plans
- Crude oil washing, inert gas, oil discharge monitor manuals
- SOPEP manual
- Details of structural fire protection

- Details of type approval for LSA, FFA and MARPOL equipment
- Details of type approval for bridge equipment and radio equipment
- Details of the company in (accordance with ISM Code) with a copy of the ISM Code
- Details of the Company Security Officer, in accordance with ISPS regulations
- Medical certificates for the crew onboard.

7.4 Certificates to be carried on board

S. No	Name of Certificate	Required under
SEC - 1	**All ships to which the referenced convention applies**	
1	International Tonnage Certificate (1969)	Tonnage Convention, 1969
2	International Load Line Certificate	LL Convention, 1966
3	International Load Line Exemption Certificate	LL Convention, 1966
4	Exemption Certificate	SOLAS 1974
5	Coating Technical File	SOLAS 1974, MSC 215(82)
6	Emergency Towing Procedure Manual	SOLAS 1974
7	Construction drawings	SOLAS 1974
8	Ship Construction File	SOLAS 1974
9	Noise Survey Report	SOLAS 1974 - Code of noise level on board ships
10	Stability information	SOLAS 1974, LL Conv, 1966
11	Damage control plans and booklets	SOLAS 1974
12	Manoeuvring booklet	SOLAS 1974
13	Evaluation of the alternative design and arrangements	SOLAS 1974
14		SOLAS 1974
15	Onboard training and drills record	SOLAS 1974
16	Fire safety training manual	SOLAS 1974
17	Fire control plan/booklet	SOLAS 1974
18	Fire safety operational booklet	SOLAS 1974
19	Operations manual for helicopter facility	SOLAS 1974
20	Statement of acceptance of the installation of replacement release and retrieval system to an existing lifeboat	SOLAS 1974
21	Muster list and emergency instructions	SOLAS 1974
22	Ship-specific Plans and Procedures for Recovery of Persons from the Water	SOLAS 1974
23	Training manual	SOLAS 1974
24	Radio record	SOLAS 1974
25	Minimum safe manning document	SOLAS 1974
26	Voyage data recorder system – certificate of compliance	SOLAS 1974
27	AIS test report	SOLAS 1974
28	Nautical charts and nautical publications	SOLAS 1974
29	LRIT conformance test report	SOLAS 1974
30	International Code of Signals and a copy of Volume III of IAMSAR Manual	SOLAS 1974

31	Records for pilot ladders used for pilot transfer	SOLAS 1974
32	Records of navigational activities	SOLAS 1974
33	Cargo Securing Manual	SOLAS 1974
34	Material Safety Data Sheets (MSDS)	SOLAS 1974
35	Safety Management Certificate	SOLAS 1974
36	Document of Compliance	SOLAS 1974
37	Continuous Synopsis Record (CSR)	SOLAS
38	Ship Security Plan and associated records	SOLAS 1974, ISPS Code
39	International Ship Security Certificate (ISSC) or Interim International Ship Security Certificate	SOLAS 1974
40	International Oil Pollution Prevention Certificate	MARPOL
41	Oil Record Book	MARPOL
42	Shipboard Oil Pollution Emergency Plan	MARPOL
43	International Sewage Pollution Prevention Certificate	MARPOL
44	Document of approval for the rate of sewage discharge	MARPOL
45	Garbage Management Plan	MARPOL
46	Garbage Record Book	MARPOL
47	International Air Pollution Prevention Certificate	MARPOL
48	International Energy Efficiency Certificate	MARPOL
49	Ozone-depleting Substances Record Book	MARPOL
50	Fuel Oil Changeover Procedure and Logbook (record of fuel changeover)	MARPOL
51	Manufacturer's Operating Manual for Incinerators	MARPOL
52	Bunker Delivery Note and Representative Sample	MARPOL
53	EEDI Technical File	MARPOL
54	Ship Energy Efficiency Management Plan (SEEMP)	MARPOL
55	Technical File	MARPOL
56	Record Book of Engine Parameters	MARPOL
57	Certificates for masters, officers or ratings	STCW 1978
58	Records of daily hours of rest	STCW 1978, MLC 2006
59	International Anti-fouling System Certificate	AFS Convention, 2001
60	Declaration on Anti-fouling System	AFS Convention, 2001
61	International Ballast Water Management Certificate	BWM 2004
62	Ballast Water Management Plan	BWM 2004
63	Ballast Water Record Book	BWM 2004
64	Certificate of insurance or other financial security in respect of civil liability for bunker oil pollution damage	BUNKER Convention 2001
65	Certificate of insurance or other financial security in respect of liability for the removal of wrecks	Nairobi WRC 2007
SEC - 2	**In addition to the certificates listed in section 1 above, passenger ships shall carry:**	
66	Passenger Ship Safety Certificate	SOLAS, 1974
67	Decision support system for masters	SOLAS, 1974
68	Search and rescue cooperation plan	SOLAS, 1974
69	List of operational limitations	SOLAS, 1974

70	Special Trade Passenger Ship Safety Certificate, Special Trade Passenger Ship Space Certificate	STP 71, SSTP 73
71	Certificate of insurance or other financial security in respect of liability for the death of and personal injury to passengers	PAL 1974 as modified by PAL PROT 2002,
SEC - 3	**In addition to the certificates listed in section 1 above, cargo ships shall carry:**	
72	Cargo Ship Safety Construction Certificate	SOLAS 1974, 1988 SOLAS PROTOCOL
73	Cargo Ship Safety Equipment Certificate	SOLAS 1974, 1988 SOLAS PROTOCOL
74	Cargo Ship Safety Radio Certificate	SOLAS 1974, 1988 SOLAS PROTOCOL
75	Cargo Ship Safety Certificate	SOLAS 1974, 1988 SOLAS PROTOCOL
76	Ship Structure Access Manual	SOLAS 1974
77	Cargo Information	SOLAS 1974
78	Bulk Carrier Booklet	SOLAS 1974, BLU Code
79	Document of authorization for the carriage of grain and grain loading manual	SOLAS 1974 & INT GRAIN CODE
80	Enhanced survey report file	SOLAS 1974
81	Dedicated Clean Ballast Tank Operation Manual	MARPOL
82	Condition Assessment Scheme (CAS) Statement of Compliance, CAS Final Report and Review Record	MARPOL
83	Subdivision and stability information	MARPOL
84	Record of oil discharge monitoring and control system for the last ballast voyage	MARPOL
85	Oil Discharge Monitoring and Control (ODMC) Operational Manual	MARPOL
86	Crude Oil Washing Operation and Equipment Manual (COW Manual)	MARPOL
87	STS Operation Plan and Records of STS Operations	MARPOL
88	VOC Management Plan	MARPOL
89	Document of approval for the stability instrument	MSC.1/Circ.1229; MSC.1/Circ.1461
90	Certificate of insurance or other financial security in respect of civil liability for oil pollution damage	CLC 1969
91	Certificate of insurance or other financial security in respect of civil liability for oil pollution damage	CLC 1992
SEC - 4	**In addition to the certificates listed in sections 1 and 3 above, where appropriate, any ship carrying noxious liquid chemical substances in bulk shall carry:**	
92	International Pollution Prevention Certificate for the Carriage of Noxious Liquid Substances in Bulk (NLS Certificate)	MARPOL
93	Cargo Record Book	MARPOL

94	Procedures and Arrangements Manual (P & A Manual)	MARPOL
95	Shipboard Marine Pollution Emergency Plan for Noxious Liquid Substances	MARPOL
SEC - 5	**In addition to the certificates listed in sections 1 and 3 above, where applicable, any chemical tanker shall carry:**	
96	Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk	BCH Code
97	International Certificate of Fitness for the Carriage of Dangerous Chemicals in Bulk	IBC Code
SEC - 6	**In addition to the certificates listed in sections 1 and 3 above, where applicable, any gas carrier shall carry:**	
98	Certificate of Fitness for the Carriage of Liquefied Gases in Bulk	IGC Code
99	International Certificate of Fitness for the Carriage of Liquefied Gases in Bulk	IGC Code
SEC - 7	**In addition to the certificates listed in sections 1, and 2 or 3 above, where applicable, any high-speed craft shall carry:**	
100	High-Speed Craft Safety Certificate	SOLAS, 1974; HSC Code, 1994
101	Permit to Operate High-Speed Craft	HSC Code, 1994
SEC - 8	**In addition to the certificates listed in sections 1, and 2 or 3 above, where applicable, any ship carrying dangerous goods shall carry:**	
102	Document of compliance with the special requirements for ships carrying dangerous goods	SOLAS 1974
SEC - 9	**In addition to the certificates listed in sections 1, and 2 or 3 above, where applicable, any ship carrying dangerous goods in packaged form shall carry:**	
103	Transport information	SOLAS 1974
104	Dangerous goods manifest or stowage plan	SOLAS 1974 MARPOL
SEC - 10	**In addition to the certificates listed in sections 1, and 2 or 3 above, where applicable, any ship carrying INF cargo shall carry:**	
105	International Certificate of Fitness for the Carriage of INF Cargo	SOLAS 1974 INF Code
SEC - 11	**In addition to the certificates listed in sections 1, and 2 or 3 above, where applicable, any Nuclear Ship shall carry:**	
106	Operating Manual for nuclear power plant	SOLAS 1974
107	A Nuclear Cargo Ship Safety Certificate or Nuclear Passenger Ship Safety Certificate, in place of the Cargo Ship Safety Certificate or Passenger Ship Safety Certificate, as appropriate.	SOLAS 1974
SEC - 12	**In addition to the certificates listed in sections 1, and 2 or 3 above, where applicable, any Ship operating in Polar waters shall carry:**	
108	Polar Ship Certificate	Polar Code
109	Polar Water Operational Manual (PWOM)	Polar Code

8.0 Certificates and Documents required on board as per ILO Convention.

S. No	Name of Certificate	Required Under
1	Maritime Labour Certificate (MLC)	MLC, 2006 Reg 5.1.3
2	Declaration of Maritime Labour Compliance Part I (DMLC I) set up by Flag State Authority	MLC, 2006 Reg 5.1.3
3	Declaration of Maritime Labour Compliance Part II (DMLC II)	MLC, 2006 Reg 5.1.3
4	Certificate of financial security for seafarers in the event of an abandonment (Standard A2.5.2) and in the event of a contractual claim involving death or long-term disability (Standard 4.2)	MLC, 2006 Reg 2.5 & 4.2 (As per 2014 Amendments to MLC, 2006)
5	Certificate of Compliance for ILO 92 (see note 1)	Accommodation of Crews (Revised)
6	Certificate of Compliance for ILO 133 (see note 1)	Accommodation of Crews (Supplementary Provisions)
7	Medical Certificate for ILO 73 (see note 1)	Fitness for Work Provisions
8	Load Test Certificate for ILO 152 (see note 2)	Occupational Safety and Health in Dock Work
9	Register Book for ILO 152 (see note 2)	Occupational Safety and Health in Dock Work

Note 1: Survey and issuance of Certificate only if required by the flag state or owners as same is coverd under MLC.

Note 2: For provision, hose handling and machinery cranes, national regulations may apply.

9.0 List of additional Statutory certificates for Indian Flag vessels

S. No	Name of Certificate	Required Under
1	Certificate of Registry	MSA
2	Certificate of Survey	MSA
3	Certificates 'A' showing the details of the passenger spaces and number of passengers that may be carried in such spaces for ships engaged in domestic voyages.	MSA
4	Certificate 'B' in respect of special trade passenger ships	MSA/STP
5	Chain register	ILO
6	Ship Sanitary Certificate	WHO
7	Medicine Chest Certificate	WHO/ILO

SOLAS CHAPTERS

CHAPTER I — General provisions


Survey


Documentation


PSC

CHAPTER II-1 — Construction - Subdivision and stability, machinery and electrical installations


Subdivision


machine/ electrical Installation


Stability

CHAPTER II -2 — Fire protection, fire detection and fire extinction


Protection


Detection


Extinction

CHAPTER III — Life-saving appliances and arrangements


Survival Craft


Personal LSA


Muster Duty

CHAPTER IV — Radiocommunications


GMDSS


Other Radio Equipment


Radio Personanel

CHAPTER V — Safety of navigation


Navigational Warning


Pilot Transfer Atrrangement


Steering Gear

learn.marineinsight.com

CHAPTER VI — Carriage of cargoes



MSDS


Loading/ Unloading/ Stowing


Grain Cargo

CHAPTER VII — Carriage of dangerous cargoes


Packaged/ Bulk Dangerous Cargo


Chemical in Bulk


Gas in Bulk

CHAPTER VIII — Nuclear ships


Nuclaer Ship Requirement

CHAPTER IX — Management for the aafe operation of ships


ISM Code


Certification


Verification

CHAPTER X — Safety measures for high-speed craft


International Code of Safety for High-Speed Craft

CHAPTER XI I & II — Special measures to enhance maritime safety & security



ESP/ CSR/


ISPS

CHAPTER XII — Additional safety measures for bulk carriers


Damage Stability/ Structural strength


Survey


Water ingress alarm

LATEST SOLAS REGULATION

CHAPTER XIII — Verification of compliance

2016
Came into force


Auditing of Contracting Government

IMO
Instrument Implementation Code

CHAPTER XIV — Polar Code

2017
Came into force


Polar Code


Ship Requirement

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